

Home Lab 4: Separating a Mixture

Do this with a parent. Goggles on. Never taste anything. Clean up after.

Goal

Take a two-powder mixture apart by dissolving and filtering, then identify both parts.

You need

From the kit: **table salt, corn starch**; iodine; cabbage indicator. From home: water, a coffee filter or paper towel, a funnel or just a cup, a spoon, a small dish, cups, paper towels. **Have your parent make the mixture** without telling you the amounts: about a spoon of salt + a spoon of corn starch, stirred together, in a cup labeled "Unknown."

Steps

1. Look at the dry "Unknown" with a magnifier if you have one. Do you see two different things? Write what you notice.
2. Add about $\frac{1}{2}$ cup of water, stir well for a minute.
3. **Filter**: pour the mixture through the coffee filter into a clean cup.
4. The stuff caught on the filter is the **residue**. The liquid that came through is the **filtrate**.
5. Scrape a bit of the wet residue onto a paper towel. Add one drop of iodine.
6. Put a few drops of the filtrate in a small dish and let it dry (or ask a parent to warm it gently). Look at what's left. Add cabbage indicator to another few drops.

Results

	Residue (on the filter)	Filtrate (the liquid)
What it looks like		
+ Iodine		
What's left when it dries		
pH (cabbage)		
Identity + formula		

Follow-up

1. Which component dissolved in the water? Which didn't?
2. Which was in the filtrate? Which was the residue?
3. Name both components with their chemical formulas.

4. If a competition sample "half dissolved" and the undissolved part turned iodine black, what would you suspect?

Coach notes

- Residue: corn starch — white, pasty, **turns iodine blue-black**. $(C_6H_{10}O_5)_n$.
- Filtrate: salt — dries to **cubic crystals**, neutral pH. NaCl.
- Follow-up: (1) salt dissolved; corn starch didn't. (2) salt in filtrate, corn starch was the residue. (3) as above. (4) a mixture of a soluble powder + a starch (corn starch or flour).